

Instructor Guide

VEXLearn Foundations Curriculum - Pilot Version

Student: _____

Date: _____

Pathway: Code Learning / Robot Programming

Team: _____

Purpose

VEXLearn Foundations helps beginners understand VEX V5 software through documented, editable code. Ask students to explain code before copying or modifying it. The curriculum prioritizes understanding, testing, and documentation over one universal performance threshold.

Recommended Schedule

Module	Focus	Suggested time	Pathway note
0	Orientation and baseline	30-60 min	Both
1	C++ fundamentals	90 min	Both
2	Project structure	60-90 min	Both
3	Configuration and control	90 min	Use hypothetical robot if needed
4	Autonomous movement	90 min	Code prediction is sufficient
5	PID laboratory	2-3 hr	Analyze code or run hardware trials
6	Reliability testing	90 min	Write a test plan or run trials
7	Autonomous capstone	2-4 hr	Written or physical submission

Completion Pathways

Code Learning Path: Modules 0-4, code-reading versions of Modules 5-7, and a written capstone. No robot is required.

Robot Programming Path: All modules, physical testing, and the autonomous capstone using a compatible tank-drive VEX V5 robot, separate left and right motor groups, a V5 inertial sensor, the official VEX VS Code extension, and a safe testing area.

Safety

- Use a clear testing area.
- Raise or restrain the robot during initial mechanism tests.
- Confirm safe mechanism states before enabling driver control.
- Use bounded movement behavior and stop testing when hardware behaves unexpectedly.
- Keep students clear of moving mechanisms and drivetrain wheels.

Teaching Guidance

- Require predictions before running code.
- Ask students to trace declarations, definitions, and calls.
- Change one PID constant at a time.
- Require battery and starting-position records during physical trials.
- Accept team-defined tolerances because robot construction differs.
- Treat unexplained compilation problems as learning evidence when students identify the likely cause.

Checkpoint Summary

Module	Completion checkpoint
0	Match at least five of six source files with their purpose
1	Score at least 75% on the module quiz
2	Explain robotConfig.cpp, robotConfig.h, and source-file relationships
3	Compile or correctly explain the remaining compilation problem
4	Explain predicted robot behavior without running the program
5	Complete eight trials and evaluate a team-defined tolerance
6	Explain why one successful run is not reliability evidence
7	Submit the complete capstone package

Assessment and Reporting

Record initial and final assessment scores, percentage-point improvement, module completion, capstone score, autonomous trial success rate, and student confidence before and after.

Do not claim educational effectiveness until results have been collected. After pilot testing, report the number of students who began and completed the course, assessment changes, capstone completion, common difficulties, and curriculum revisions made from feedback.

Instructor Notes
